basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

NATIONAL SENIOR CERTIFICATE
NASIONALE SENIOR SERTIFIKAAT

GRADE/*GRAAD* 11

MATHEMATICS P2/*WISKUNDE V2*

NOVEMBER 2015

MEMORANDUM

MARKS/*PUNTE*: 150

This memorandum consists of 17 pages.
Hierdie memorandum bestaan uit 17 bladsye.

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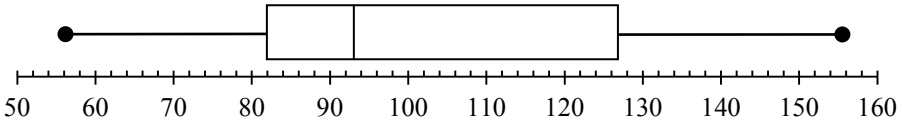
NOTE:
- If a candidate answered a question TWICE, mark only the FIRST attempt.
- If a candidate crossed out an answer and did not redo it, mark the crossed-out answer.
- Consistent accuracy applies to ALL aspects of the marking memorandum.
- Assuming values/answers in order to solve a problem is unacceptable.

LET WEL:
- *As 'n kandidaat 'n vraag TWEE keer beantwoord het, sien slegs die EERSTE poging na.*
- *As 'n kandidaat 'n antwoord deurgehaal en nie oorgedoen het nie, sien die deurgehaalde antwoord na.*
- *Volgehoue akkuraatheid is op ALLE aspekte van die memorandum van toepassing.*
- *Dit is onaanvaarbaar om waardes/antwoorde aan te neem om 'n probleem op te los.*

QUESTION/*VRAAG* 1

56	68	69	71	71	72	82	84	85
88	89	90	92	93	94	96	97	99
102	103	127	128	134	135	137	144	156

1.1	Range/*Omvang* = 156 – 56 = 100 kg	✓ max – min ✓ answer/*antw* (2)
1.2	Mode/*Modus* = 71 kg	✓ answer/*antw* (1)
1.3	Median/*Mediaan* = T_{14} = 93 kg	✓ answer/*antw* (1)
1.4	$Q_1 = T_7 = 82$ $Q_3 = T_{21} = 127$ IQR = $Q_3 - Q_1$ = 127 – 82 = 45 kg	✓ $Q_1 = 82$ ✓ $Q_3 = 127$ ✓ answer/*antw* (3)
1.5		✓ box/*mond* ✓ whiskers/*snor* (2)
1.6	SD = 25,838 ≈ 25,84 kg	✓✓ answer/*antw* (2)
1.7	$\bar{x}$ = 98,59 $\bar{x} + 1\sigma$ = 98,59 + 25,84 = 124,43 kg 127 > 124,43 ∴ I agree with this person/*Ek stem met die persoon saam*.	✓ $\bar{x}$ = 98,59 ✓ 124,43 ✓ conclusion/*gevolgtrekking* (3) **[14]**

QUESTION/*VRAAG* 2

WEIGHT LOSS OVER 4 WEEKS ***GEWIGSVERLIES IN 4 WEKE*** **(IN GRAMS/*GRAM*)**	**FREQUENCY** ***FREKWENSIE*** ***f***	**CUMULATIVE** ***KUMULATIEWE*** ***f***
$1\,000 < x \leq 1\,500$	2	2
$1\,500 < x \leq 2\,000$	3	5
$2\,000 < x \leq 2\,500$	3	8
$2\,500 < x \leq 3\,000$	4	12
$3\,000 < x \leq 3\,500$	5	17
$3\,500 < x \leq 4\,000$	7	24
$4\,000 < x \leq 4\,500$	2	26
$4\,500 < x \leq 5\,000$	1	27

2.1

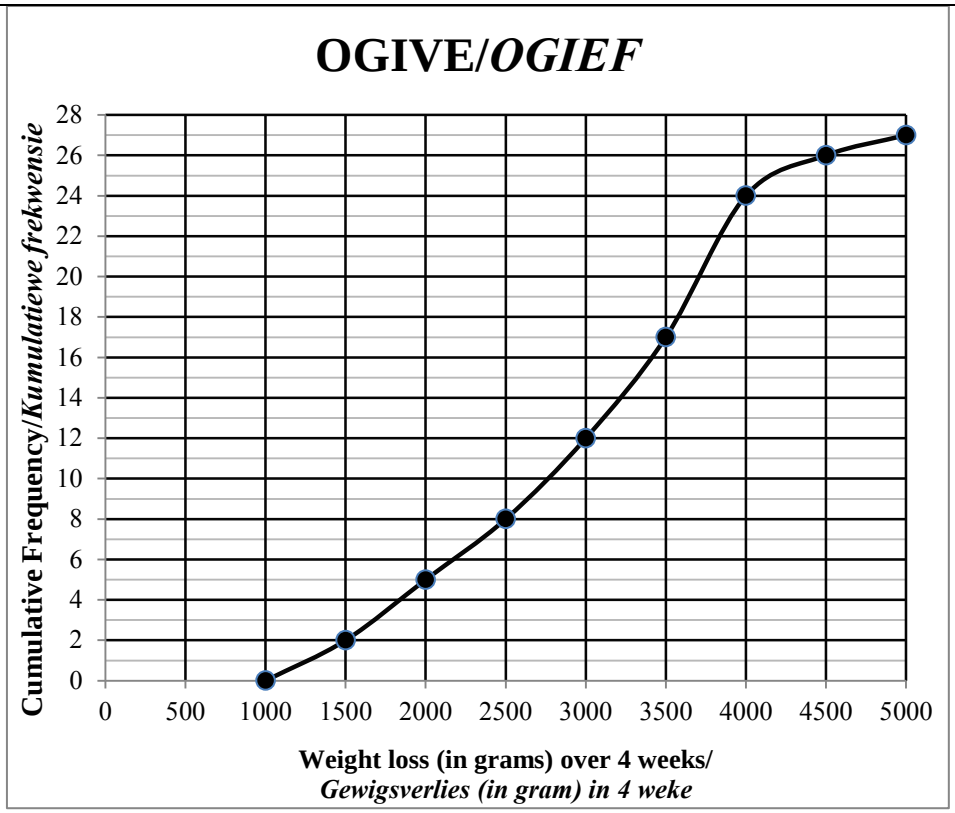
Average/*Gemiddelde* $= \dfrac{1250\times 2+1750\times 3+\ \ldots.\ \ +4750\times 1}{27}$

$= \dfrac{81250}{27}$

$= 3009{,}259 \approx 3009{,}26$ g

✓ 81 250
✓ answer/*antw* (2)

2.2



✓ grounded
✓ upper limits
✓ cumulative frequency
✓ smooth curve

✓ *geanker*
✓ *boonste limiete*
✓ *kumulatiewe frekwensie*
✓ *gladde kurwe* (4)

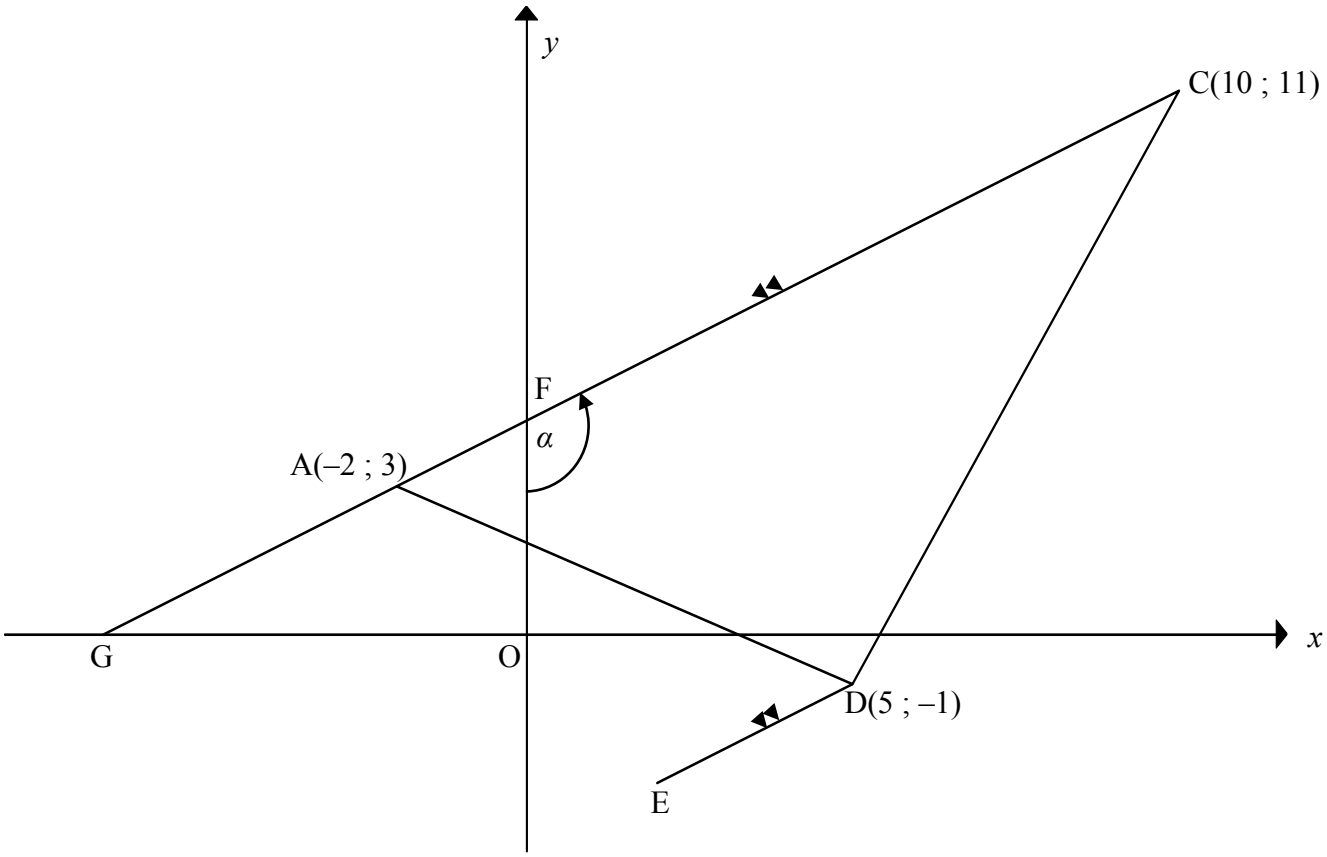
2.3

(3200 ; 14)
27 – 14 = 13 participants/*deelnemers* (accept/*aanvaar*: 12 – 14)

✓ 14
✓ 13 (2)

[8]

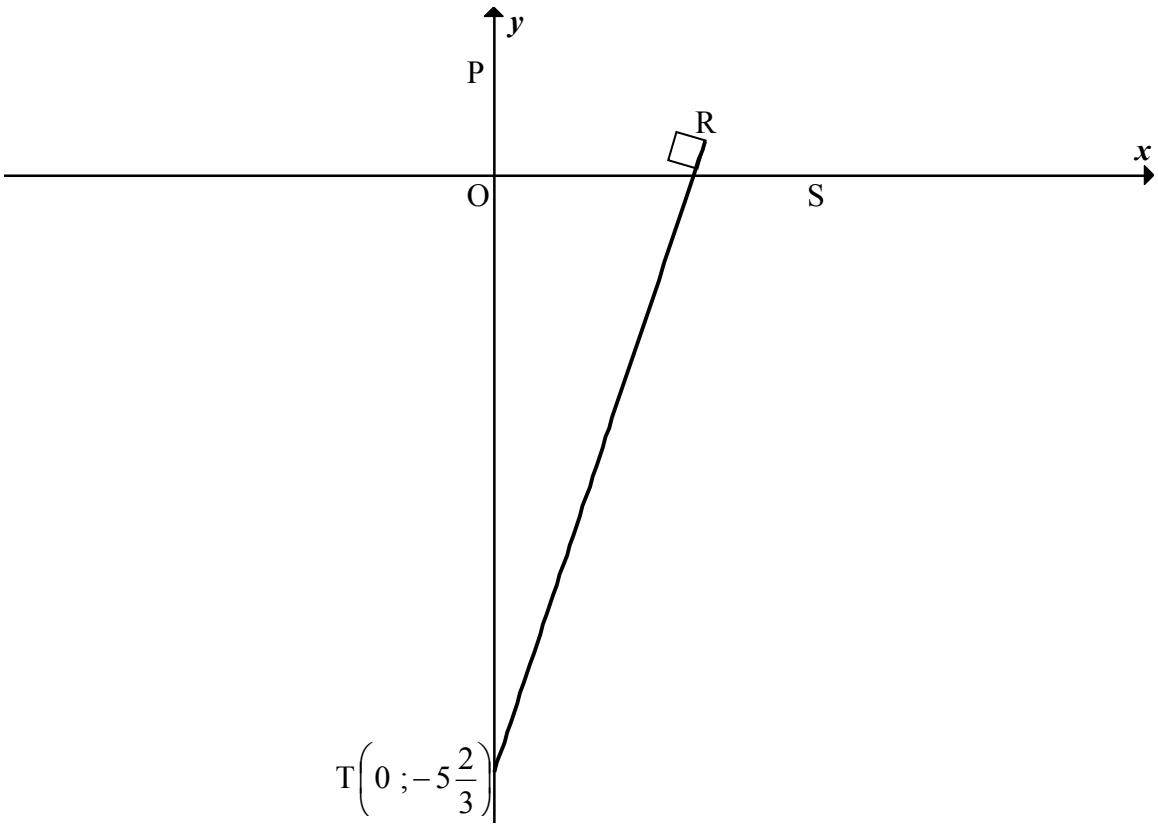
QUESTION/*VRAAG* 3



3.1	$m_{AC} = \frac{11-3}{10-(-2)}$ $= \frac{8}{12} = \frac{2}{3}$	✓ subst into/*in* gradient form/ *gradiëntvorm* ✓ answer/*antw* (2)
3.2	$m_{DE} = m_{AC} = \frac{2}{3}$ [DE | | CA] $y - y_1 = m(x - x_1)$ $\quad y = mx + c$ $y-(-1) = \frac{2}{3}(x-5)$ $\quad -1 = \frac{2}{3}(5) + c$ $y = \frac{2}{3}x - \frac{13}{3}$ **OR/*OF*** $-\frac{13}{3} = c$ $y = \frac{2}{3}x - \frac{13}{3}$	✓ gradients equal/ *gradiënte gelyk* ✓ subst m & (5 ; –1) into eq of straight line/ *in vgl v rt lyn* ✓ equation/*vgl* (3)
3.3	$\tan F\hat{G}O = m_{AC} = \frac{2}{3}$ $F\hat{G}O = 33{,}69°$ $\therefore \alpha = 90° + 33{,}69°$ [ext/*buite* ∠ of Δ] $= 123{,}69°°$	✓ $\tan F\hat{G}O = m_{AC}$ ✓ 33,69° ✓ answer/*antw* (3)

3.4.1	midpoint of BE = midpoint of AD [diagonals of rectangle] *midpt v BE = midpt v AD [hoeklyne van reghoek]* $M\left(\frac{-2+5}{2};\frac{3+(-1)}{2}\right)$ $= M\left(\frac{3}{2};1\right)$	✓ R ✓ *x*-value/*waarde* ✓ *y*-value/*waarde* (3)
3.4.2	BE = AD [diagonals of rect /*hoeklyne van regh*] BE = $\sqrt{(-2-5)^2+(3-(-1))^2}$ $= \sqrt{49+16}$ $= \sqrt{65}$ units/*eenh*	✓ R ✓ subst into distance formula/*subst in afstandsformule* ✓ answer/*antw* (3) **[14]**

QUESTION/*VRAAG* 4

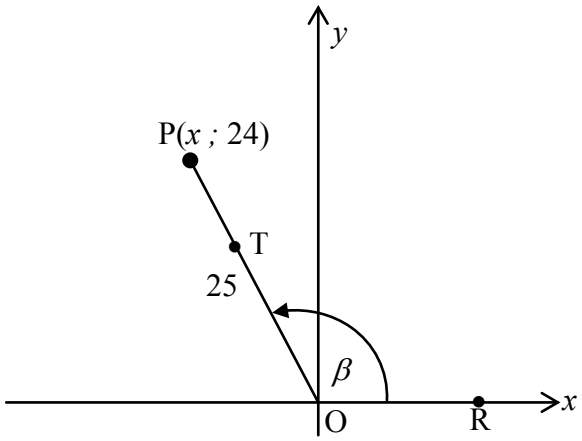


4.1	$x + ay - a = 0$ $0 + ay - a = 0$ $y = 1$ $\therefore$ P(0 ; 1) **OR/*OF*** $x + ay - a = 0$ $y = -\frac{x}{a} + 1$ $\therefore$ P(0 ; 1)	✓ $x = 0$ ✓ $y = 1$ (2) ✓ $x = 0$ ✓ $y = 1$ (2)
4.2	OP = 1 unit/*eenh* $\therefore$ SP = 3 units/*eenh* $\therefore$ S(3 ; 0) $3 + a(0) - a = 0$ $\therefore a = 3$ **OR/*OF*** $m_{PS} = -\frac{PO}{OS} = -\frac{1}{3}$ $y = -\frac{1}{a}x + 1$ $-\frac{1}{a} = -\frac{1}{3}$ $\therefore a = 3$	✓ S(3 ; 0) ✓ answer/*antw* (2) ✓ $m_{PS} = -\frac{1}{3}$ ✓ answer/*antw* (2)

4.3	$m_{PS} = -\frac{PO}{OS} = -\frac{1}{3}$ $m_{PS} \times m_{TR} = -1$ (RT $\perp$ PS) $\therefore m_{TR} = 3$ $\therefore y = 3x - 5\frac{2}{3}$	✓ $m_{PS} = -\frac{1}{3}$ ✓ $m_{TR} = 3$ ✓ equation/*vgl* (3)
4.4	$x + 3y = 3$ and/*en* $y = 3x - 5\frac{2}{3}$ $x + 3(3x - 5\frac{2}{3}) = 3$ $x + 9x - 17 = 3$ $10x = 20$ $x = 2$ $\therefore y = 3(2) - 5\frac{2}{3} = \frac{1}{3}$ $\therefore$ R$(2 ; \frac{1}{3})$	✓ subst of eq of TR into/*subst v vgl v TR in* $x + 3y = 3$ ✓ simplify/*vereenv* ✓ $x = 2$ ✓ $y = \frac{1}{3}$ (4)
4.5	Area of/*Opp van* $\Delta PRT = \frac{1}{2} PT \times \perp h$ $= \frac{1}{2}(6\frac{2}{3})(2)$ $= 6\frac{2}{3}$ sq units/*vk eenh*	✓ $6\frac{2}{3}$ ✓ $\perp h$ ✓ answer/*antw* (3)
4.6	PT is the diameter of the circle passing through P, R and T. [PT subtends 90° at circumference] *PT is die middellyn vd sirkel wat deur P, R en T gaan. [PT onderspan 90° op omtrek]* $r = \frac{1}{2} \times 6\frac{2}{3}$ $= 3\frac{1}{3}$	✓ S/R ✓ answer/*antw* (2) **[16]**

QUESTION/*VRAAG* 5

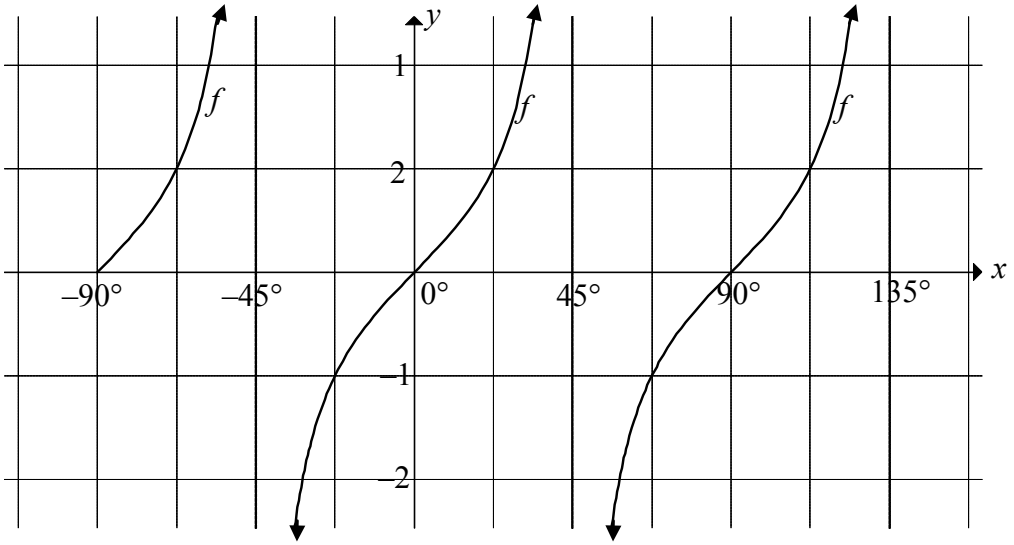
5.1

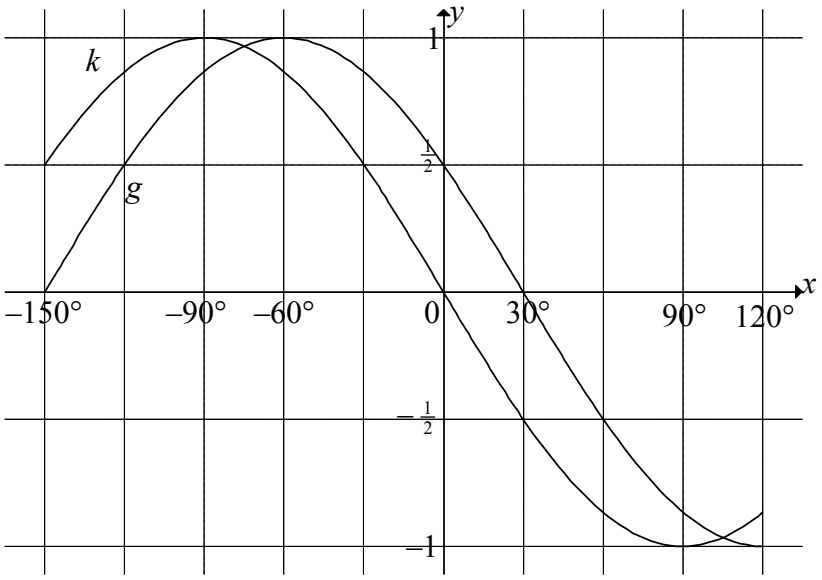


5.1.1	$x^2 + y^2 = r^2$ $x^2 + (24)^2 = 25^2$ $x^2 = 49$ $x = -7$	✓ subst ✓ answer/*antw* (2)
5.1.2(a)	$\sin\beta = \frac{24}{25}$	✓ answer/*antw* (1)
5.1.2(b)	$\cos(180° - \beta)$ $= -\cos\beta$ $= -(-\frac{7}{25})$ $= \frac{7}{25}$	✓ reduction/*reduksie* ✓ answer/*antw* (2)
5.1.2(c)	$\tan(-\beta)$ $= -\tan\beta$ $= -(\frac{24}{-7})$ $= \frac{24}{7}$	✓ reduction/*reduksie* ✓ answer/*antw* (2)
5.1.3	$\sin\beta = \frac{24}{25} = \frac{y_T}{15}$ $\therefore y_T = 14\frac{2}{5}$ $\cos\beta = -\frac{7}{25} = \frac{x_T}{15}$ $\therefore x_T = -4\frac{1}{5}$ $\therefore T(-4\frac{1}{5} ; 14\frac{2}{5})$	✓ equating ratios/ *stel verh's gelyk* ✓ y_T ✓ equating ratios/ *stel verh's gelyk* ✓ x_T (4)

5.2	$\dfrac{2\sin x.\cos x.(1+\dfrac{\sin^2 x}{\cos^2 x})}{\dfrac{\sin x}{\cos x}}$ $= 2\sin x.\cos x.(\dfrac{\cos^2 x+\sin^2 x}{\cos^2 x})\times\dfrac{\cos x}{\sin x}$ $= 2\cos^2 x(\dfrac{1}{\cos^2 x})$ $= 2$ **OR/*OF*** $\dfrac{2\sin x.\cos x.(1+\dfrac{\sin^2 x}{\cos^2 x})}{\dfrac{\sin x}{\cos x}}$ $= 2\sin x.\cos x.(1+\dfrac{\sin^2 x}{\cos^2 x})\times\dfrac{\cos x}{\sin x}$ $= 2\cos^2 x(1+\dfrac{\sin^2 x}{\cos^2 x})$ $= 2(\cos^2 x+\sin^2 x)$ $= 2$	✓ $\tan x = \dfrac{\sin x}{\cos x}$ ✓ identity/*identiteit* ✓ simplify/*vereenv* ✓ answer/*antw* (4) ✓ $\tan x = \dfrac{\sin x}{\cos x}$ ✓ simplify/*vereenv* ✓ identity/*identiteit* ✓ answer/*antw* (4)
5.3.1	$\dfrac{1-\cos^2 A}{4(-\sin A)}$ $= \dfrac{\sin^2 A}{-4\sin A}$ $= -\dfrac{1}{4}\sin A$	✓ $-\sin A$ ✓ identity/*identiteit* ✓ answer/*antw* (3)
5.3.2	$-\dfrac{1}{4}\sin 2x = 0{,}21$ $\sin 2x = -0{,}84$ ref/*verw* $\angle = 57.14°$ $\therefore\ 2x = 237{,}14° + k.360°$ or $2x = 302{,}86° + k.360°$ $x = 118{,}57° + k.180°$ or $x = 151{,}43° + k.180°$; $k \in Z$	✓ correct equation/ *korrekte vgl* ✓ $-0{,}84$ ✓ ref/*verw* $\angle$ $= 57{,}14°$ ✓ $237{,}14°$ & $302{,}86°$ ✓ $118{,}57°$ & $151{,}43°$ ✓ $k.180°$; $k \in Z$ (6) **[24]**

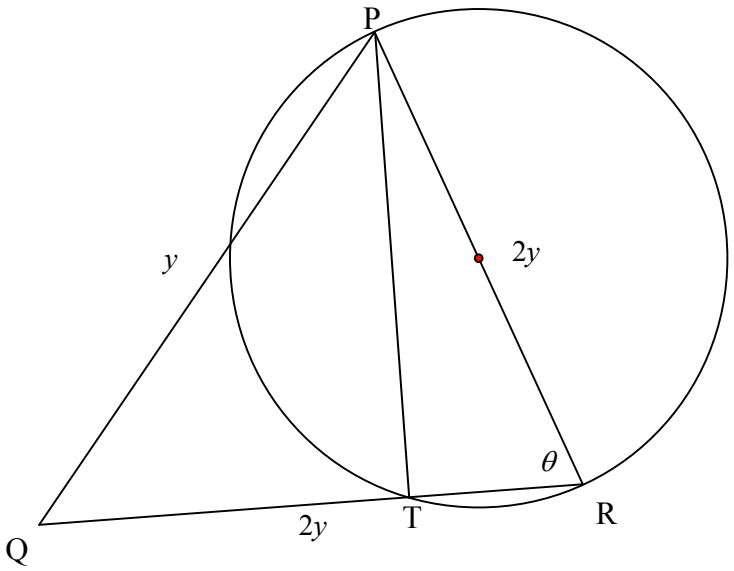
QUESTION/*VRAAG* 6



6.1.1	$b = 2$	✓ answer/*antw* (1)
6.1.2	$45° < x \leq 67{,}5°$ **OR/*OF*** $x \in (45° ; 67{,}5°]$	✓ extreme values/ *grenswaardes* ✓ notation/*notasie* (2)
6.1.3	The graph of f has to shift 55° to the left to form the graph of h. *Die grafiek f het 55° na links geskuif om grafiek h te vorm.* $x = –10°$ and/*en* $x = 80°$	✓ $x = –10°$ ✓ $x = 80°$ (2)
6.2.1		✓ TP(–90° ; 1) ✓ TP(90° ; -1) ✓ $(–150° ; \frac{1}{2})$ and/*en* (120° ; –0,87) ✓ y-intercept/*afsnit* (4)

6.2.2	min value/*waarde* $= -1 - 3 = -4$	✓✓ answer/*antw* (2)
6.2.3	$\cos(x + 60°) = -\sin x$ $= \cos(90° + x)$ $x + 60° = -(90° + x)$ *or* $x + 60° = 90° + x$ $2x = -150° + k.360°$ no solution $x = -75° + k.180°$ $\therefore x = -75°$ or/*of* $105°$	✓ co-ratio/*ko-verh* ✓✓ correct equations *korrekte vgls* ✓ $x = -75° + k.180°$ ✓ $-75°$ ✓ $105°$ (6)
	$\cos(x + 60°) = -\sin x$ $= -\cos(90° - x)$ $x + 60° = 180° - (90° - x)$ *or* $x + 60° = 180° + (90° - x)$ $x = 90° + x + k.360°$ $2x = 210° + k.360°$ no solution/*geen oplossing* $x = 105° + k.180°$ $\therefore x = -75°$ or $105°$	✓ co-ratio/*ko-verh* ✓✓ correct equations *korrekte vgls* ✓ $x = 105° + k.180°$ ✓ $-75°$ ✓ $105°$ (6)
	OR/*OF* $\cos(x + 60°) = -\sin x$ $\sin[90° - (x + 60°)] = -\sin x$ $150° - x = 180° + x$ *or* $150° - x = 360° - x$ $-2x = 30° + k.360°$ no solution/*geen oplossing* $x = -15° - k.180°$ $\therefore x = -75°$ or $105°$	✓ co-ratio/*ko-verh* ✓✓ correct equations *korrekte vgls* ✓ $x = -15° - k.180°$ ✓ $-75°$ ✓ $105°$ (6)
6.2.4	$-75° < x < 105°$ **OR/*OF*** $x \in (-75° ; 105°)$	✓ extreme values/ *grenswaardes* ✓ notation/*notasie* (2)
6.2.5	$y = \cos(x + 60°)$ $= \sin[90° - (x + 60°)]$ $= \sin(30° - x)$ $= -\sin(x - 30°)$ $\therefore \theta = 30°$ **OR/*OF*** By inspections/*Deur inspeksie*: $\theta = 30°$	✓ co-ratio/*ko-verh* ✓ answer/*antw* (2) ✓✓ answer/*antw* (2) **[21]**

QUESTION/*VRAAG* 7



7.1	$PQ^2 = QR^2 + PR^2 - 2QR.PR.\cos\theta$ $y^2 = (2y)^2 + (2y)^2 - 2(2y)(2y).\cos\theta$ $\cos\theta = \frac{4y^2 + 4y^2 - y^2}{2(2y)(2y)}$ $= \frac{7y^2}{8y^2}$ $= \frac{7}{8}$	✓ subst into cosine rule/*in cos-reël* ✓ cos θ as subject/ *as onderwerp* ✓ simplify/*vereenv* ✓ answer/*antw* (4)
7.2	PT ⊥ QR [∠ in semi-circle/*halfsirkel*] $\sin\theta = \frac{PT}{PR} = \frac{PT}{2y}$ $\therefore$ PT = 2*y* sin θ **OR/*OF*** PT ⊥ QR [∠ in semi-circle/*halfsirkel*] $\hat{Q} = 90° - \frac{\theta}{2}$ [∠s opposite = sides/*∠e to = sye*] $\sin(90° - \frac{\theta}{2}) = \frac{PT}{y}$ $PT = y.\cos\frac{\theta}{2}$	✓ R ✓ correct ratio/ *korrekte verh* ✓ answer/*antw* (3) ✓ R ✓ correct ratio/ *korrekte verh* ✓ answer/*antw* (3 **[7]**

QUESTION/VRAAG 8

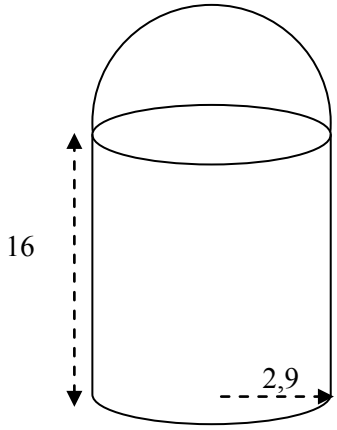


FIGURE 1

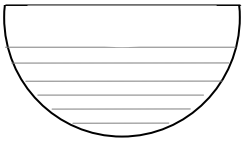
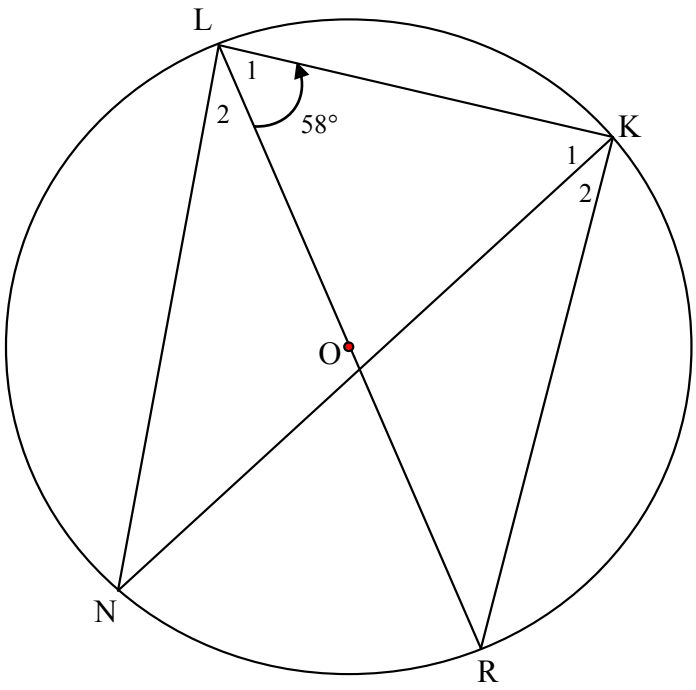


FIGURE 2

8.1	Surface area(SA) $= 2\pi rh + \pi r^2 + 2\pi r^2$ *Buite-opp* $= 2\pi rh + 3\pi r^2$ $= 2\pi(2,9)(16) + 3\pi(2,9)^2$ $= 118,03\pi$ $= 370,80 \text{ cm}^2$	✓ SA of tin/*OA v blik* ✓ SA of hemisphere/ *BO v hemisfeer* ✓ SA of base/ *BO v basis* ✓ subst ✓ answer/*antw* (5)
8.2	Volume of hemisphere/*v hemisfeer* $= \frac{1}{2}\times\frac{4}{3}\pi r^3$ $= \frac{1}{2}\times\frac{4}{3}\pi(2,9)^3$ $= 16,26\pi$ 80% of/*van* volume $= 0,8 \times 16,26\pi$ $= 40,86 \text{ cm}^3$ **OR/*OF*** Volume of hemisphere/*v hemisfeer* $= \frac{1}{2}\times\frac{4}{3}\pi r^3$ 80% of/*van* volume $= 0,8 \times \frac{1}{2}\times\frac{4}{3}\pi r^3$ $= 0,8 \times \frac{1}{2}\times\frac{4}{3}\pi(2,9)^3$ $= 40,86 \text{ cm}^3$	✓ V of hemisphere ✓ subst ✓ answer/*antw* (3) ✓ vol of hemisphere/ *vol van hemisfeer* ✓ subs ✓ answer/*antw* (3) **[8]**

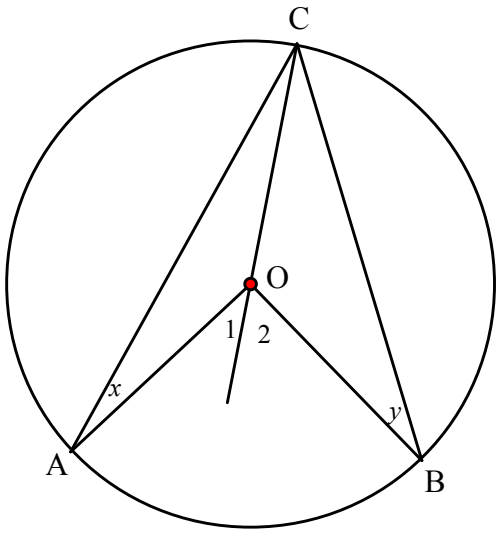
QUESTION/*VRAAG* 9



9.1	$L\hat{K}R = 90°$ [∠ in semi-circle/*halfsirkel*]	✓ S ✓ R (2)
9.2	$\hat{R} = 180° - (90° + 58°) = 32°$ [∠s/*e* of/*van* Δ]	✓ S ✓ R (2)
9.3	$\hat{N} = 32°$ [∠s/*e* in same segment/*dieselfde segment*]	✓ S ✓ R (2) **[6]**

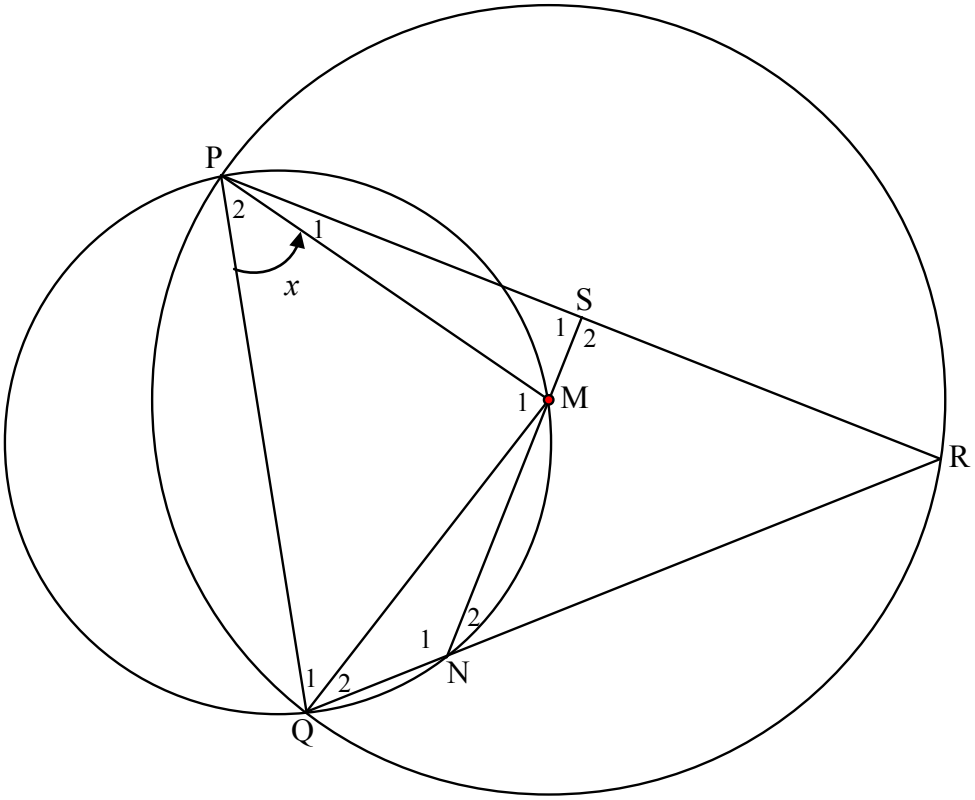
QUESTION/*VRAAG* 10

10.1



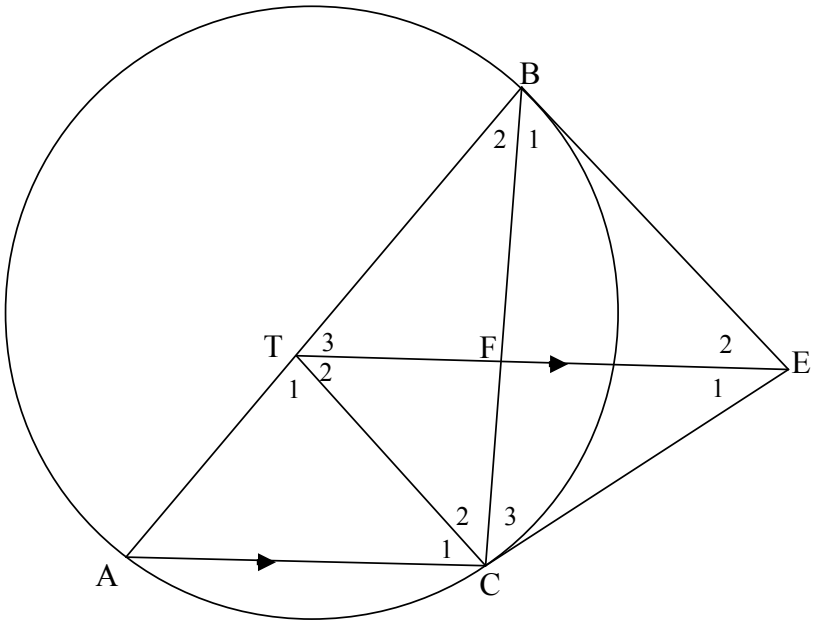
10.1.1	AO and/*en* CO are radii/*is radiusse* $\hat{A} = \hat{C}_1 = x$ [∠s opp equal sides/*∠e to gelyke sye*] $\hat{O}_1 = \hat{A} + \hat{C}_1 = 2x$ [ext/*buite* ∠ of/*van* Δ]	✓ S ✓ S ✓R (3)
10.1.2	$\hat{B} = \hat{C}_2 = y$ [∠s opp equal sides/*∠e to gelyke sye*] $\hat{O}_2 = \hat{B} + \hat{C}_2 = 2y$ [ext/*buite* ∠ of/*van* Δ] $A\hat{O}B = 2x + 2y$ $= 2(x + y)$ $= 2(\hat{C}_1 + \hat{C}_2)$ $= 2A\hat{C}B$	✓S ✓S ✓S (3)

10.2



10.2.1	ext ∠ of cyc quad/*buite ∠ v koordevh*	✓ R (1)
10.2.2	MP = QM [radii] $\hat{Q}_1 = x$ [∠s opp equal sides/*∠e to gelyke sye*]	✓ S ✓ R (2)
10.2.3	$\hat{M}_1 = 180° - 2x$ [∠s/*e* of/*van* Δ] $\hat{R} = 90° - x$ [∠ at centre = 2× ∠ at circumference/ *midpts∠ = 2 × omtreks∠*]]	✓ S ✓ S ✓ R (3)
10.2.4	In ΔNSR: $\hat{R} = 90° - x$ and $\hat{N}_2 = x$ $\hat{S}_2 = 180° - (90° - x + x)$ [∠s/*e* of/*van* Δ] $= 90°$ PS = SR [line from centre ⊥ chord/*lyn v midpt ⊥ kd*]	✓ S ✓ S ✓ R (3) **[15]**

QUESTION/*VRAAG* 11



11.1	$\hat{B}_1 = \hat{A}$ [tangent-chord theorem/*rkl-kdst*] $\hat{A} = \hat{T}_3$ [corresp ∠s/*ooeenk ∠e* ; TE | | AC] $\therefore \hat{B}_1 = \hat{T}_3$	✓ S ✓ R ✓ S ✓ R (4)
11.2	BE = CE [tangents from same point/*rklyne v dieselfde pt*] $\hat{B}_1 = \hat{C}_3$ [∠s opp equal sides/*∠e to gelyke sye*] $\hat{C}_3 = \hat{T}_3$ [$\hat{B}_1 = \hat{T}_3$] ∴TBEC a cyclic quad [converse ∠s in the same segment] *TBEC is koordevh* [*omgekeerde ∠e in dieselfde segment*]	✓ R ✓ S ✓ S ✓ R (4)
11.3	$\hat{B}_1 = \hat{T}_2$ [∠s in the same segment/*∠e in dieselfde segment*] $\hat{B}_1 = \hat{T}_3$ [proven/*bewys* in 11.1] $\therefore \hat{T}_2 = \hat{T}_3$ ∴ ET bisects/*halveer* $B\hat{T}C$	✓ S ✓ R (2)
11.4	$\hat{B}_2 = \hat{E}_2$ [tangent-chord theorem/*rkl-kdst*] $\hat{C}_2 = \hat{E}_2$ [∠s in the same segment/*∠e in dieselfde segment*] ∴ TB = TC [sides opposite equal ∠s/*sye to gelyke ∠e*]	✓ S ✓ R ✓ R ✓ R (4)
11.5	$\hat{C}_1 = \hat{T}_2$ [alternate/*verwiss* ∠s/*e* ; TE | | AC] $\therefore \hat{C}_1 = \hat{A}$ ∴ AT =TC [sides opposite equal ∠s/*sye to gelyke ∠e*] T is a point that is equidistant from A, B and C on the circle/ *T is 'n punt wat ewever is vanaf A,B en C op die sirkel*. ∴ T is the centre of the circle/*T is die middelpunt van die sirkel*.	✓ S ✓ S ✓ S (3) **[17]**

TOTAL/*TOTAAL*: 150

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